

CHAPTER-2

REQUIREMENT ANALYSIS

2.1 Requirement Analysis

Authentication

- **Student Login** – Students can access their personal account by entering their registered login credentials.
- **Student Logout** – Students can securely log out of the system after completing their activities.
- **Invalid Login** – If incorrect login credentials are entered, the system will prevent the student from accessing the account.
- **Admin Login** – The administrator can access administrative functions using authorized credentials.

Process data

- **Display** – Students can view their dashboard, tasks, timetable, notes, study plans, goals, habits, and academic information. The administrator can view the required student-related information available in the system.
- **Edit** – Students can edit their personal profile, tasks, schedules, notes, goals, and other information according to the available features.
- **Search** – Students can search for required notes, resources, tasks, and other information available in the system.
- **Update** – Students can update their tasks, schedules, notes, goals, habits, and other personal information whenever required.

Task & To-Do Management

- **Add Task** – Students can create tasks and add the required details such as task name, deadline, and other relevant information.
- **Manage Task** – Students can view, update, and manage their tasks according to their academic and personal requirements.

Student Registration

- **Add New Student** – New students can register in the system by providing the required personal information. The student details are stored in the database and an account is created for accessing StudentHub.

Notes & Resource Management

- **Add Notes** – Students can create and store academic notes and study-related resources.
- **Manage Resources** – Students can view, edit, organize, and manage their notes and available study resources.

Habit Tracker

- **Add Habit** – Students can create habits that they want to develop or maintain.
- **Track Habit** – Students can record their habit activities and monitor their consistency over time.

Study Planner

- **Study Planning** – Students can create and manage study plans according to their academic requirements.
- **Study Tracking** – Students can monitor their planned study activities and progress.

Timetable & Class Schedule

- **Add Schedule** – Students can add their class and academic schedules to the system.
- **Manage Schedule** – Students can view and update their timetable and class-related information whenever required.

Performance Report

- **Performance Report** – The system can provide a performance report based on the student's academic activities, tasks, goals, and progress information.

Finance & Expense Tracker

- **Add Expense** – Students can record their daily or academic-related expenses in the system.
- **Expense Tracking** – Students can view and monitor their expenses to manage their finances effectively.

Mood & Mental Health Journal

- **Mood Entry** – Students can record their mood and personal well-being information.
- **Journal Management** – Students can maintain personal journal entries to monitor their emotional well-being over time.

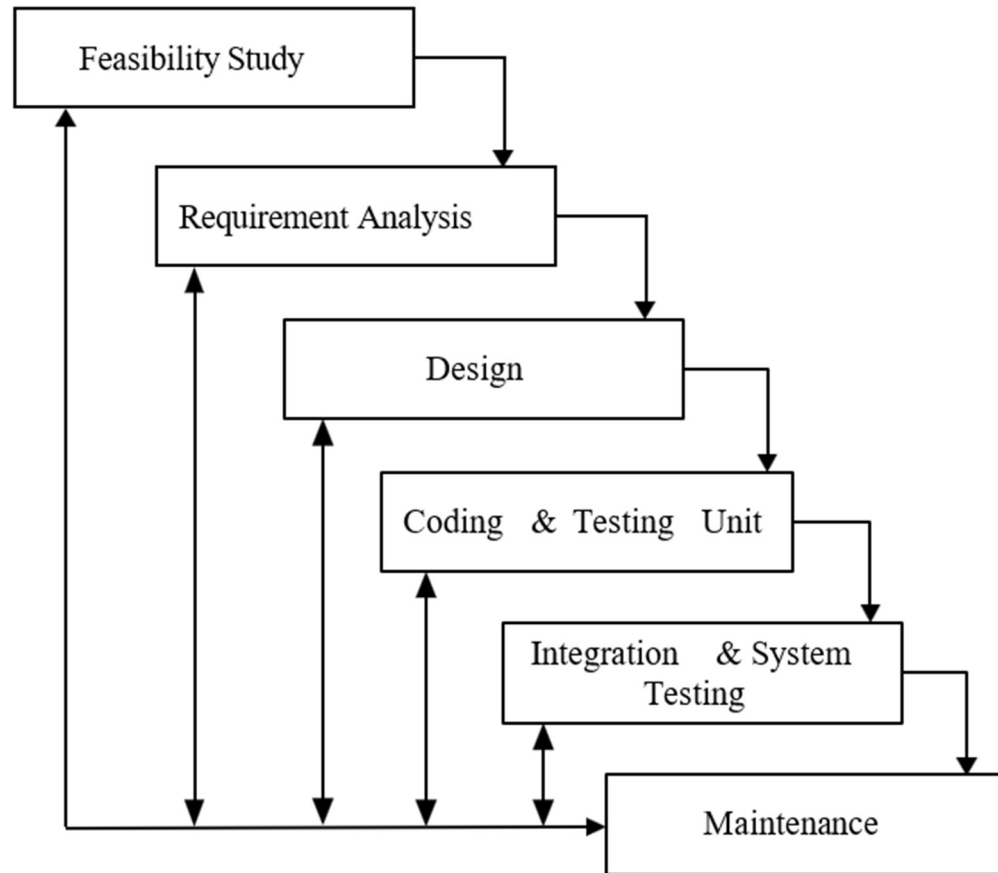
Goal & Milestone Management

- **Managing Goals** – Students can create and manage their academic or personal goals and divide them into milestones to track their progress.

Data Export & Privacy Settings

- **Data Export** – Students can export their available personal or academic information when required.
- **Privacy Settings** – Students can manage the available privacy .

2.2 Project Model



[Figure 1: Iterative Waterfall Model]

- The StudentHub application is developed using the **Iterative Waterfall Model**. The development process is divided into different phases such as feasibility study, requirement analysis, system design, implementation, testing, and maintenance.
- In this model, the system is developed through a sequence of phases. Each phase is completed systematically, and the results can be reviewed and improved during subsequent iterations.

- The system is divided into different modules that can be developed and implemented systematically.
- The initial version provides the basic functionalities required for student life management.
- Additional functionalities can be added and existing features can be improved in successive iterations.
- Errors can be identified and corrected during the development and testing phases.
- The system can be refined through successive versions by improving existing functionalities and adding new requirements.

Advantages

- Each successive version performs more useful work than the previous version.
- The core modules are tested thoroughly, thereby reducing the chance of errors in the final product.
- The model is more flexible and less costly to change the scope and requirements.
- This model helps in fulfilling exact user requirements.
- Feedback provided at each increment is useful for determining a better final product.
- The iterative approach allows improvements to be made at each stage of the development process.

2.3 Schedule Representation

Project scheduling helps in organizing different development activities according to their start and finish dates. The major activities involved in the development of the system are requirement analysis, system analysis, system design, system coding, and testing and integration.

[Table 1: Schedule Representation]

ACTIVITY	START DATE	FINISH DATE
Requirement Analysis		
System Analysis		
System Design		
System Coding		
Testing and Integration		

2.4 Feasibility Study

2.4.1 Technical Feasibility

- The proposed system will be developed as a web-based application and requires appropriate web technologies. The technologies required to develop the StudentHub system are available.
- Currently available web technologies can be used for developing the StudentHub application.
- Servers – Apache.
- DBMS – MySQL

2.4.2 Economical Feasibility

- Economic feasibility determines whether the benefits of the proposed StudentHub system justify the development and operational costs. The system reduces manual effort and provides a convenient platform for managing academic and personal activities. If the expected benefits outweigh the costs, the system is considered economically feasible.

2.4.3 Operational Feasibility

- It is the measure of how well the proposed StudentHub system solves the problems identified during the problem analysis phase and satisfies the requirements identified during requirement analysis.
- Potential users of the system can easily understand the website navigation and handling. Hence, training up to the necessary level would be easy.
- The implementation of the system through a web-based platform can be easily managed. The system provides a simple interface that allows students to manage their tasks, schedules, notes, goals, and other activities conveniently.
- Securing the data by providing appropriate access levels for students and administrators. User authentication and authorization will help maintain secure access to the system.
- Student, academic, task, schedule, and other personal information can be easily stored and retrieved through the system, making student life management more convenient and efficient.